

Experiment - 6.1.2. Factorial of a Number

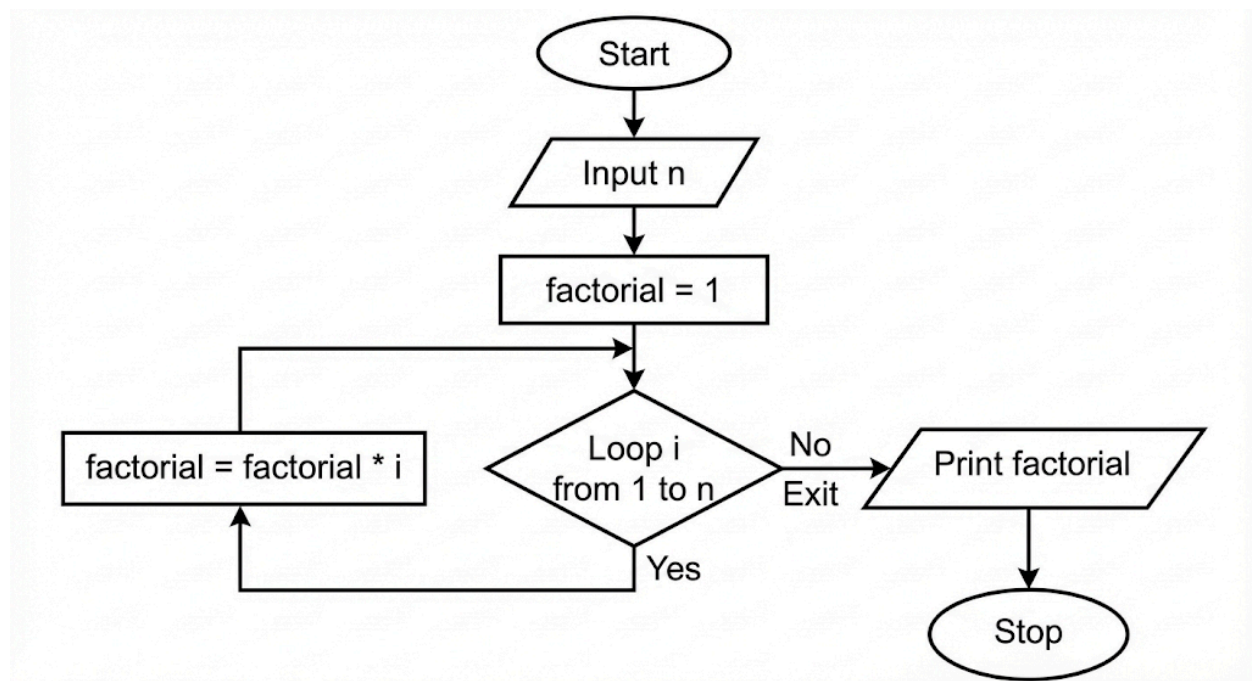
1. Aim

To design and implement a Python program that calculates the factorial of a given integer n using a for loop. The program reads the integer, computes its factorial iteratively, and prints the final result.

2. Pseudocode

1. **START**
2. **READ** a single integer from the user and **STORE** it in the variable n .
3. **INITIALIZE** a variable factorial to 1.
4. **FOR** each number i in the range from 1 to n (inclusive):
 - **MULTIPLY** factorial by i (factorial = factorial * i).
 - **STORE** the new value back into factorial.
5. **END FOR** loop once all numbers up to n have been multiplied.
6. **PRINT** the final value of factorial.
7. **END**

3. Flowchart



4. Python Program

```
n = int(input())
```

```
factorial = 1
```

```
for i in range(1, n + 1):  
    factorial *= i
```

```
print(factorial)
```

5. Experiment Screenshot

The screenshot displays the CDETANTRA online IDE interface. On the left, the problem statement for '6.1.2. Factorial of a Number' is shown, including input and output formats. The main editor on the right contains the Python code for calculating the factorial of a number n using a loop. Below the code editor, the execution results are displayed, showing that the program passed all test cases and the output for a sample input of 10 is 3628800.

Problem Statement: 6.1.2. Factorial of a Number

Write a Python program to calculate the factorial of a number n using loops.

Input Format:

- A single line containing an integer n .

Output Format:

- Print the factorial of the given integer n .

Sample Test Cases

Code:

```
1 n = int(input())  
2  
3 factorial = 1  
4  
5 for i in range(1, n + 1):  
6     factorial *= i  
7  
8 print(factorial)
```

Execution Results:

- Average time: 0.003 s (3.25 ms)
- Maximum time: 0.004 s (4.00 ms)
- 2 out of 2 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

Test case 1:

Expected output	Actual output
3628800	3628800